

Syllabus

Spring semester 2026

Course Title	General Physics 2	Course Code		EPE3103		Credit	4	Classroom	Hyung 3204
		Type		자연계열/교과심화		Hours	4		
Teacher	Name	Class no.	Office	Office Hour*	Contact		E-mail		
	Lee, Hun Bong	1	Chang 5505		Anytime(e-mail)		superpig@ksa.hs.kr		

*Office Hour: Time for students to ask questions and discuss lecture or lab topics with teachers for deeper understanding.

1. Course Descriptions

General Physics 2 is one-semester course which covers Electricity and Magnetism, Electric Circuits, Electromagnetic Waves and Wave Optics, Relativity and Quantum theory. The course is a university level course providing a deep understanding of the concepts and principles of physics and emphasizing the creative thinking. The course is offered for students who have completed General Physics I course and who have completed or are enrolled concurrently in a differential and integral calculus course. A systematic approach to problem solving is considered as a basic way to get better understanding. Questions and problems also develop students' mathematical skill.

2. Core Target Competencies

Subject Core Competencies			
Critical Thinking Skills	Analytical and Creative Thinking	Complex Problem-Solving Skills	Creativity
Fundamental concepts in several subjects in physics will be presented, including electromagnetism, optics, and subjects in modern physics, are introduced. Following the homework and the discussion sessions in this lecture, students will be encouraged to develop their competencies listed above. The ways of thinking, trained in this lecture, will provide important background for future coursework.			

3. Textbooks and References

- A. Textbook: University Physics with Modern Physics, 2nd Ed. (Bauer, Westfall, McGraw-Hill)
- B. References
 - The Feynman Lectures on Physics I, II, III (R. Feynman et al., Addison-Wesley)
 - Berkeley Physics Course I (McGraw-Hill)

4. Class Method

- A. Self-learning during homework, assessment through short quizzes.
- B. The lectures will provide students with various approaches to the subject that differ from the textbook.

C. Wide use of physical demonstrations and computer simulations.

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5. Evaluation

A. Evaluation Criteria

Midterm	Final	Quizzes	Homework	Attendance and Attitude	Total
33	34	0	17	16	100

B. Evaluation Method

- (1) The midterm and final exams are given through written tests.-
- (2) Quizzes will be given several times during the semester.-
- (3) Homework: Penalty will be given to the overdue homework.-
- (4) Attendance: -0.3 point per late & -1.0 point per absence without notice.

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6. Criteria for Awarding Credits

☞ Pass: In the case of receiving a **score of 80 or more** out of 100 according to the criteria of the evaluation method.

☞ **Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal's Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.**

- Attendance for at least three-quarters of the total class hours is required to earn credit.
- For Pass/Fail courses, if the instructor determines that a specific evaluation element is insufficient and requires a separate evaluation, a separate evaluation may be conducted.
- ● Calculation of the basis score for awarding credit to absentees :
- ● If a basis score cannot be calculated, the grading method :
- ※ Refer to the "Academic Performance Management Regulations" for the percentage of credit points applied.

7. Weekly Plans

Week (period)	Topic and Contents	(Notes)
Week 1 2.23.~3.1.	Ch.21 Electrostatics Ch.22 Electric Fields and Gauss's Law	Report would be represented after finishing each chapter
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 2 3.2.~3.8.	Ch.23 Electric Potential	3.2. Substitute Holiday
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 3 3.9.~3.15.	Ch.24 Capacitors	
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 4 3.16.~3.22.	Ch.25 Current and Resistance Ch.26 Direct Current Circuits	
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 5 3.23.~3.29.	Ch.27 Magnetism Ch.28 Magnetic Fields of Moving Charges	
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 6 3.30.~4.5.	Ch.29 Electromagnetic Induction	
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 7 4.6.~4.12.	Ch.30 Alternating Current Circuits	
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 8 4.13.~4.19.	Midterm Exams (4.14.~4.17.)	4.13. Regular Classes

Week 9 4.20.~4.26.	Ch.31 Electromagnetic Waves	
Week 10 4.27.~5.3.	Ch.34 Wave Optics	
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 11 5.4.~5.10.	Ch.35 Relativity	5.5.
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	Children's Day
Week 12 5.11.~5.17.	Ch.36 Quantum Physics	
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 13 5.18.~5.24.	Ch.37 Quantum Mechanics	
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	
Week 14 5.25.~5.31.	Ch.38 Atomic Physics	5.25.
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	Substitute Holiday
Week 15 6.1.~6.7.	Week for a missing lecture	6.3.
Core Competencies	Critical Thinking Skills, Analytical and Creative Thinking, Complex Problem-Solving Skills, Creativity	Local Election Day
Week 16 6.8.~6.14.	Make-up class week	
Week 17 6.15.~6.19.	Final Exams (6.18.~6.19.)	

※ The above plans may change according to the academic schedule.

[Reference] Syllabus Competence Reference

1. Analytical and Creative Thinking	2. Complex Communication	3. Leadership and Teamwork	4. Digital/ Quantitative Literacy	5. Global Perspective	6. Adaptability/ Progression / Spirit of Adventure	7. Integrity and Ethical Decision Making	8 Habits/ Mindset
Ability to identify, manage and deal with complex problems	Ability to understand and express themselves in two or more languages	Ability to be the first to come up with and present new ideas	Ability to understand, use and apply digital technology	Ability to develop an open mind about the traditions or values of others	Ability to develop flexibility, agility and adaptability	Ability to sustain an empathetic and understanding perspective	Sincerity
Ability to see through prejudice and identify reliable information and uncertain information	Ability to communicate clearly with diverse audiences or readers	Ability to influence people to follow themselves, rather than lead them through directives, orders, or authority	Ability to create digital contents and media	Ability to understand the history, politics, religion, and culture of another region	Ability to challenge unfamiliar situations	Ability to cultivate sincerity, honesty, fairness, and respect creativity	Creativity
Ability to find quality information in the flood of information	Ability to listen attentively	Ability to build trust, resolve conflicts, and support others	Ability to utilize multimedia resources to effectively communicate one's ideas in a variety of formats	Ability to use one or more of the international languages used worldwide	Ability to explore and experiment	Ability to display moral courage in the face of injustice	A love of learning and curiosity
Ability to articulate meaningful questions	Ability to speak effectively and coherently	Ability to facilitate group discussions or discussions, build consensus, and negotiate outcomes	Ability to master and use higher mathematics	Ability to use technology to connect people and things around the world	Ability to word effectively in situations of ambiguous or changing priorities	Ability to act responsibly for the benefit and happiness of the greater community	Resilience/elasti city
Ability to analyze and create ideas or knowledge	Ability to write concisely and clearly, considering the diversity of readers	Ability to teach, guide, and counsel Ability to seek help from others	Ability to understand traditional or recently emerging fields such as mathematics, science, technology, environmental science, robotics, geometry, cellular automata, nanotechnology, and biotechnology	Social and intellectual skills to effectively navigate information across multiple cultures	Ability to see failure as a lesson and to accept that innovation is made up of frequent failures and small successes	Ability to have an intrinsic understanding of ethical issues or dilemmas about new media or new technologies	Perseverance/pa tience
Ability to devise and test solutions to problems by	Ability to explain information and persuasively	Ability to collaborate on tasks, manage groups, and		Ability to use 21 st century core competencies such as communication	Ability to develop a sense of independence to seek new roles,	Ability to make rational and ethical decisions on complex	Self-efficacy

1. Analytical and Creative Thinking	2. Complex Communication	3. Leadership and Teamwork	4. Digital/ Quantitative Literacy	5. Global Perspective	6. Adaptability/ Progression / Spirit of Adventure	7. Integrity and Ethical Decision Making	8 Habits/ Mindset
trial and error	engage others in an engaging way	delegate responsibilities		skills, collaboration skills, critical thinking skills, creativity to properly understand and deal with global issues	ideas, or strategies	issues	
Ability to think of different ways to cope		Ability to listen and practice goals		Ability to learn from and collaborate with people from different cultures, religions and lifestyle through mutual respect and open dialogue	Ability to develop the knowledge and competencies necessary to start a business and run a business		Stress management skills
Ability to develop knowledge and perspectives that are not limited to one field but span multiple fields		Ability to share credit for success		Ability to generate diverse ideas and exploit social and cultural differences to get things done			Time management skills
Ability to make sustained inferences							
Ability to put things together or adapt them to new uses and situations							
Ability to solve new problems that cannot be solved through established rules							
Ability to use knowledge or creativity to solve complex real-world problems							

1. Complex Problem-Solving Skills	Ability to solve new and unclear problems that may arise in complex real-world situations
2. Critical Thinking Skills	Ability to compare and analyze the strengths and weaknesses of various solutions or approaches to a given problem using logic and reasoning
3. Creativity	Ability to come up with novel ideas when given a certain topic or situation or to suggest a creative method in the process of solving a problem
4. Human Resource Management Skills	Ability to select the most suitable person for a specific task and manage them to achieve maximum work performance through motivation, and systematically efficient management
5. Ability to Collaborate	Ability to adjust one's own behavior in response to the behavior of others
6. Emotional Ability	Ability to perceive other people's reactions and understand why they react the way they do
7. Judgement and Decision-making Skills	Ability to make the most rational judgement and decision by analyzing the costs and benefits of a specific situation or problem
8. Orientated Towards Service	Ability to actively seek ways to help others
9. Negotiation Skills	Ability to bring together parties who differ in ideas or interests and try to reduce those differences
10. Cognitive Flexibility	Ability to apply different kinds of rules or principles to rearrange and reorganize things in different ways